

Markov Decision Processes

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1 : MDP

MDP (fully observable) . “ ” Markov , (reward) (action) .

$$\underbrace{\langle S, P \rangle}_{\text{Markov Process}} \xrightarrow{+R, \gamma} \underbrace{\langle S, P, R, \gamma \rangle}_{\text{MRP}} \xrightarrow{+A} \underbrace{\langle S, A, P, R, \gamma \rangle}_{\text{MDP}}$$



MDP π MRP , MRP Markov Process . (evaluation) “ MRP ” .

2 Markov Process

2.1 Markov

$$\mathbb{P}[S_{t+1} | S_t] = \mathbb{P}[S_{t+1} | S_1, \dots, S_t]$$

S_t (sufficient statistic) . history .

2.2

$$P_{ss'} = \mathbb{P}[S_{t+1} = s' | S_t = s], \quad \sum_{s'} P_{ss'} = 1 \quad (\quad = 1)$$

Markov Process (Chain) $\langle S, P \rangle$ — S P . .

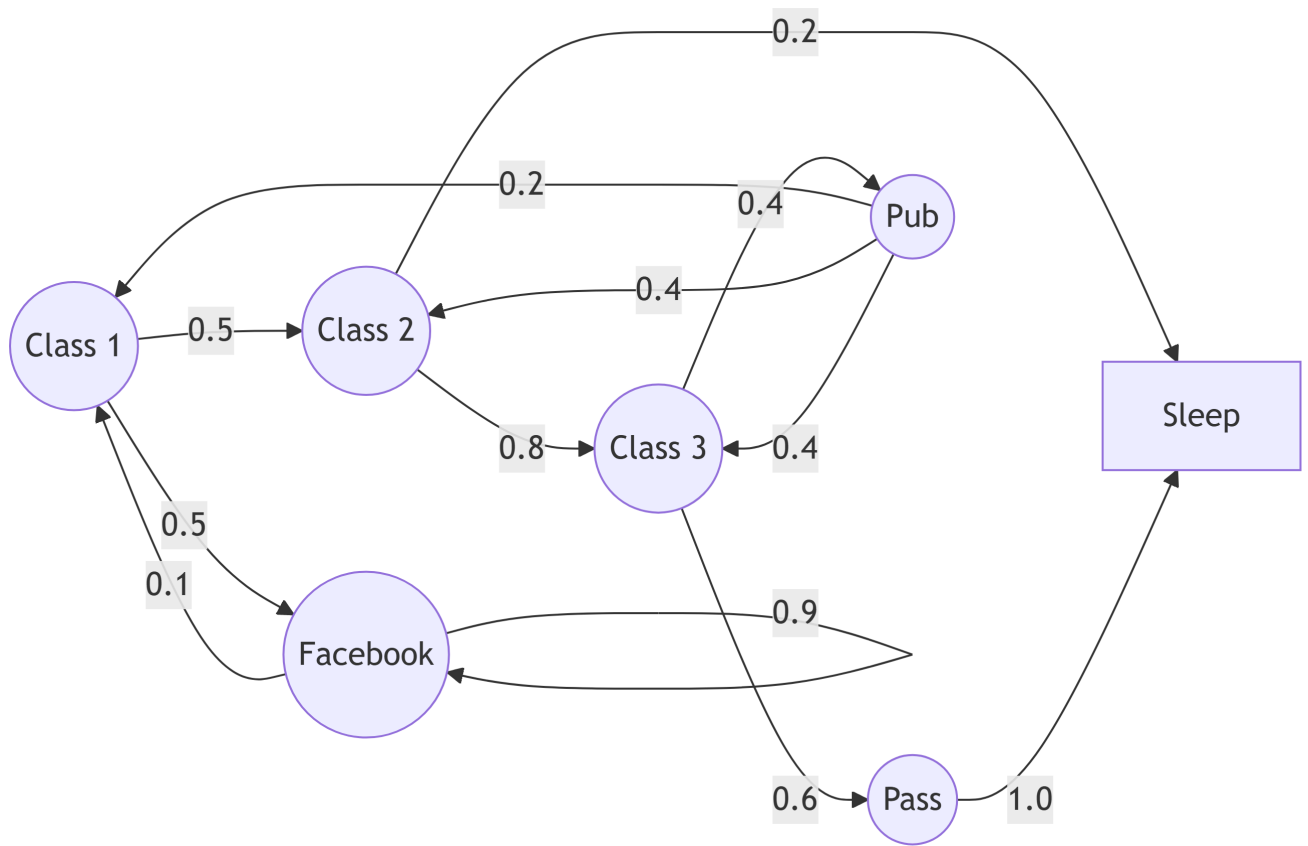
2.3 Student Markov Chain

(sample episode) : C1 $\rightarrow$ C2 $\rightarrow$ C3 $\rightarrow$ Pass $\rightarrow$ Sleep, C1 $\rightarrow$ FB $\rightarrow$ FB $\rightarrow$ C1 $\rightarrow$ C2 $\rightarrow$ Sleep .

3 Markov Reward Process (MRP)

MRP = $\langle S, P, R, \gamma \rangle$ — Markov chain (value) .

- : $R_s = \mathbb{E}[R_{t+1} | S_t = s]$
- : $\gamma \in [0, 1]$



1: Student Markov Chain

3.1 Return ()

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

i

$$s \quad R_{t+1} \cdot G_t \quad , \quad v(s) \quad .$$

3.2 — γ

- ()
- (cyclic) Markov process **return**
-
-
- /
- (terminate) $\gamma = 1$ (undiscounted)

3.3 Bellman Equation

$$v(s) = \mathbb{E}[G_t \mid S_t = s]$$

Return + .

$$\begin{aligned} v(s) &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma v(S_{t+1}) \mid S_t = s] \\ &= R_s + \gamma \sum_{s'} P_{ss'} v(s') \end{aligned}$$

$$\mathbb{E}[G_{t+1} \mid S_{t+1} = s'] = v(s') \quad \text{tower property} \quad .$$

3.4

$$v = R + \gamma P v \implies (I - \gamma P) v = R \implies v = (I - \gamma P)^{-1} R$$

- , $O(n^3)$ ($n = |S|$).
- $\gamma < 1 \implies \rho(\gamma P) < 1 \implies (I - \gamma P)^{-1}$. n DP / Monte-Carlo / TD .

3.5 γ

Sleep (terminal, $v = 0$) $\{C1, C2, C3, \text{Pass}, \text{Pub}, FB\}$ $v_T = (I - \gamma Q)^{-1} R_T$.
 Q , $\gamma = 1 \implies \rho(Q) < 1$.

```

import numpy as np
import matplotlib.pyplot as plt

states = ["C1", "C2", "C3", "Pass", "Pub", "FB"]
idx = {s: i for i, s in enumerate(states)}

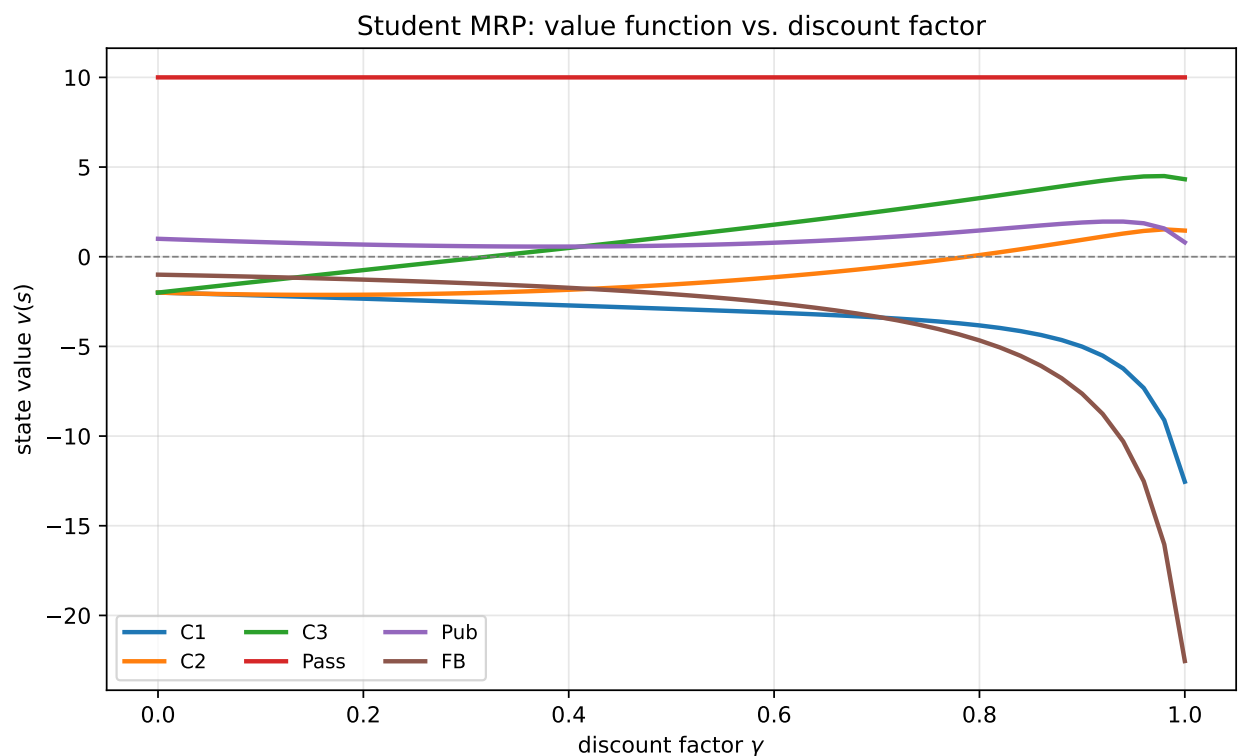
Q = np.zeros((6, 6))
Q[idx["C1"], idx["FB"]] = 0.5; Q[idx["C1"], idx["C2"]] = 0.5
Q[idx["C2"], idx["C3"]] = 0.8 # C2 -> Sleep 0.2 (terminal)
Q[idx["C3"], idx["Pub"]] = 0.4; Q[idx["C3"], idx["Pass"]] = 0.6
# Pass -> Sleep 1.0 (terminal): 0
Q[idx["Pub"], idx["C1"]] = 0.2
Q[idx["Pub"], idx["C2"]] = 0.4
Q[idx["Pub"], idx["C3"]] = 0.4
Q[idx["FB"], idx["FB"]] = 0.9; Q[idx["FB"], idx["C1"]] = 0.1

R = np.array([-2, -2, -2, 10, 1, -1], dtype=float)

gammas = np.linspace(0.0, 1.0, 51)
V = np.array([np.linalg.solve(np.eye(6) - g * Q, R) for g in gammas])

fig, ax = plt.subplots(figsize=(8, 5))
for j, s in enumerate(states):
    ax.plot(gammas, V[:, j], marker="", linewidth=2, label=s)
ax.axhline(0, color="gray", linewidth=0.8, linestyle="--")
ax.set_xlabel(r"discount factor $\gamma$")
ax.set_ylabel(r"state value $v(s)$")
ax.set_title("Student MRP: value function vs. discount factor")
ax.legend(ncol=3, fontsize=9)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()

```



2: gamma v(s)



$\gamma = 0$ $v(s) = R_s (\quad) \cdot \gamma$, $R = +10$ Pass C3, C2 . 0.9
 FB γ $R = -1$ ($\gamma = 1$ $v(FB) \approx -22.5$).

γ	C1	C2	C3	Pass	Pub	FB
0.0	-2.00	-2.00	-2.00	+10.00	+1.00	-1.00
0.5	-2.91	-1.55	+1.12	+10.00	+0.62	-2.08
0.9	-5.01	+0.94	+4.09	+10.00	+1.91	-7.64
1.0	-12.54	+1.46	+4.32	+10.00	+0.80	-22.54

4 Markov Decision Process (MDP)

MDP = $\langle S, A, P, R, \gamma \rangle$ — MRP (decision) .

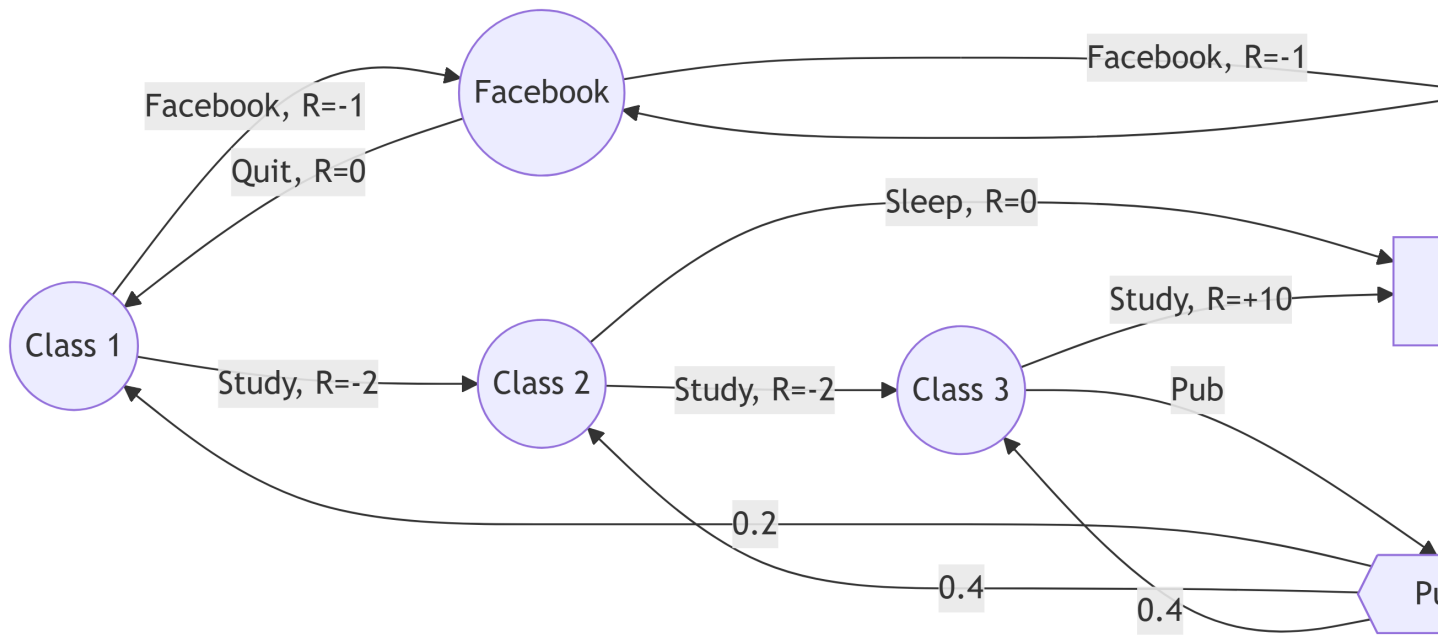
$$P_{ss'}^a = \mathbb{P}[S_{t+1} = s' \mid S_t = s, A_t = a], \quad R_s^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$$

4.1 (policy)

$$\pi(a \mid s) = \mathbb{P}[A_t = a \mid S_t = s]$$

(stationary), . history Markov .

4.2 Student MDP



3: Student MDP (= action / reward)

i

MRP “ ” , MDP “ ” . Pub (0.2/0.4/0.4) , .

4.3 → MRP

π $S_1, S_2, \dots$ Markov process $\langle S, P^\pi \rangle$, - MRP $\langle S, P^\pi, R^\pi, \gamma \rangle$.

$$P_{ss'}^\pi = \sum_a \pi(a \mid s) P_{ss'}^a, \quad R_s^\pi = \sum_a \pi(a \mid s) R_s^a$$

4.4

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \quad q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

v_π , q_π (,) .

4.5 Bellman Expectation Equation

v q .

$$\begin{aligned} v_{\pi}(s) &= \sum_a \pi(a | s) q_{\pi}(s, a) \\ q_{\pi}(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a v_{\pi}(s') \\ v_{\pi}(s) &= \sum_a \pi(a | s) \left(R_s^a + \gamma \sum_{s'} P_{ss'}^a v_{\pi}(s') \right) \\ q_{\pi}(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a \sum_{a'} \pi(a' | s') q_{\pi}(s', a') \end{aligned}$$

:

$$v_{\pi} = R^{\pi} + \gamma P^{\pi} v_{\pi} \implies v_{\pi} = (I - \gamma P^{\pi})^{-1} R^{\pi}$$

MRP .

5 (Optimality)

5.1

$$v_*(s) = \max_{\pi} v_{\pi}(s), \quad q_*(s, a) = \max_{\pi} q_{\pi}(s, a)$$

MDP “ ” v_* (q_*) .

5.2

MDP .

1. π_* .
2. $v_{\pi_*}(s) = v_*(s)$.
3. $q_{\pi_*}(s, a) = q_*(s, a)$.

5.3

q_* greedy (deterministic) .

$$\pi_*(a | s) = \begin{cases} 1 & \text{if } a = \arg \max_{a \in A} q_*(s, a) \\ 0 & \text{otherwise} \end{cases}$$



• q_* MDP •

5.4 Bellman Optimality Equation

$$\begin{aligned} v_*(s) &= \max_a q_*(s, a) \\ q_*(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a v_*(s') \\ v_*(s) &= \max_a \left(R_s^a + \gamma \sum_{s'} P_{ss'}^a v_*(s') \right) \\ q_*(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a \max_{a'} q_*(s', a') \end{aligned}$$

5.5 Expectation vs. Optimality

Bellman Expectation	$\sum_a \pi(a s) (\cdot)$	$(I - \gamma P^\pi)^{-1} R^\pi$
Bellman Optimality	$\max_a (\cdot)$	$\rightarrow$

max , Value Iteration, Policy Iteration, Q-learning, Sarsa •

6 Extensions to MDPs

6.1 • MDP

- (countably infinite) / :
- / : -2 (LQR)
- : → **HJB(Hamilton-Jacobi-Bellman)** , Bellman time-step $\rightarrow 0$

6.2 POMDP ()

$\langle S, A, O, P, R, Z, \gamma \rangle$. MDP = HMM.

- **Belief state** $b(h)$: history h
- POMDP () **history tree** () **belief state tree**

6.3 Undiscounted / MDP

- **Ergodic** Markov process = recurrent() + aperiodic()
 - π chain ergodic MDP ergodic
 - ergodic MDP ρ^π
 - **average-reward Bellman equation** (differential value function)
-

7



1. $v = (I - \gamma P)^{-1} R$ — Bellman . (MRP/leaky bucket .)
2. **Expectation vs. Optimality** 4 — $v \leftrightarrow q$. max vs. $\sum_a \pi$ / .
3. γ 6 .
4. $+ q_*$ greedy .
5. $\rightarrow$ **MRP** (P^π, R^π) — .

leaky bucket · contention “Markov chain ” , .